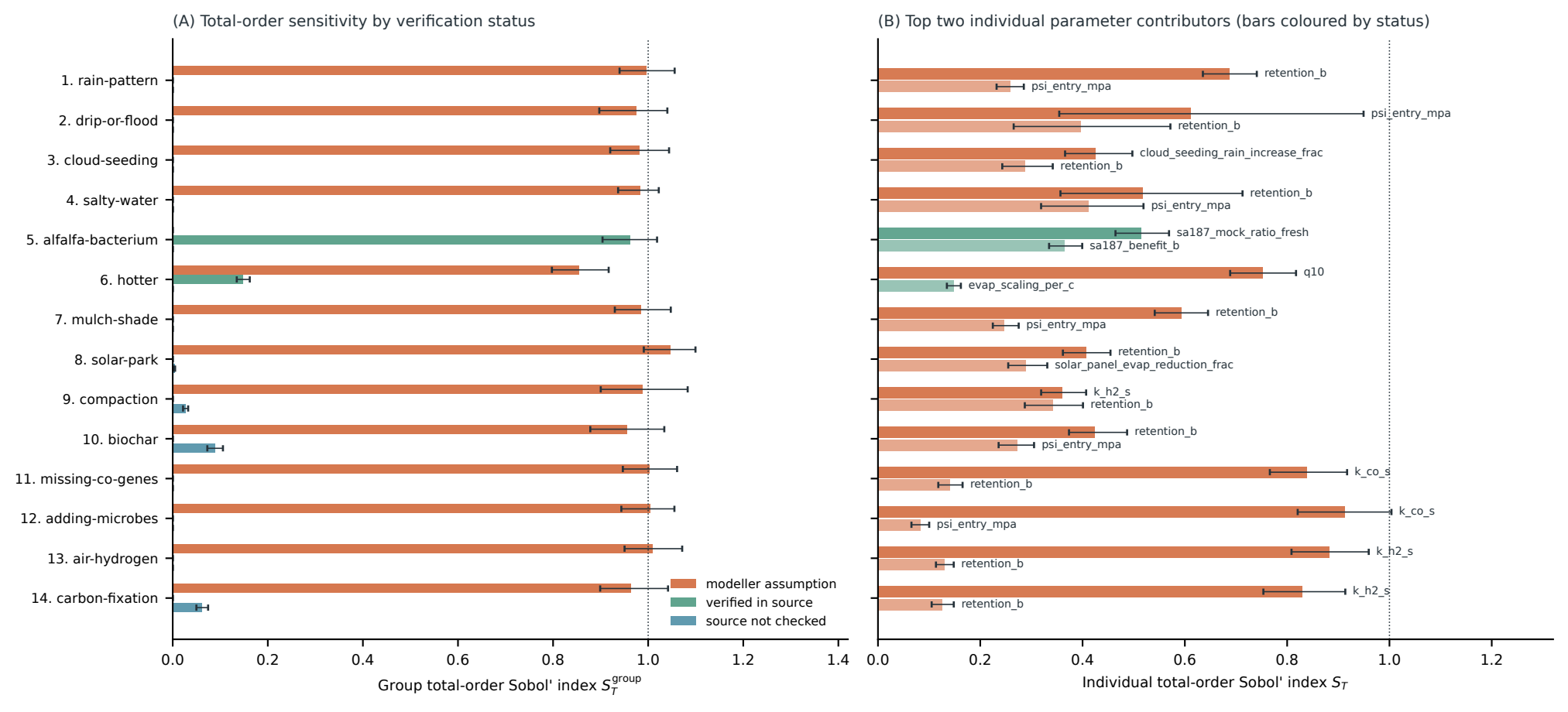
(A) Total-order sensitivity by verification status
1. rain-pattern
2. drip-or-flood
3. cloud-seeding
4. salty-water
5. alfalfa-bacterium
6. hotter
7. mulch-shade
8. solar-park
9. compaction
10. biochar
11. missing-co-genes
12. adding-microbes
13. air-hydrogen
14. carbon-fixation
modeller assumption
verified in source
source not checked
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Group total-order Sobol' index S_T^{group}
(B) Top two individual parameter contributors (bars coloured by status)
retention_b
psi_entry_mpa
psi_entry_mpa
retention_b
cloud_seeding_rain_increase_frac
retention_b
retention_b
psi_entry_mpa
sa187_mock_ratio_fresh
sa187_benefit_b
q10
evap_scaling_per_c
retention_b
psi_entry_mpa
retention_b
solar_panel_evap_reduction_frac
k_h2_s
retention_b
retention_b
psi_entry_mpa
k_co_s
retention_b
k_co_s
psi_entry_mpa
k_h2_s
retention_b
k_h2_s
retention_b
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Individual total-order Sobol' index S_T